Building the Proton from Quarks via Cobordism Merges

session review — the bipartite obstruction, the trivalent junction, and the road to the color singlet

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Abstract

This note reviews a single working session on the #380 epic — building a proton from quarks with the canonical C++ `MergeCobordism` primitive, reading *emergent* observables off the relaxed geometry rather than imposing the answer. Two issues were carried to a definite state. (#382) A *bipartite* merge provably cannot produce the color singlet, because the singlet is the three-index alternating invariant ε_{ijk}, absent from $\mathbf{3} \otimes \mathbf{3}$; the register merge was refactored onto the exact period read-out and the result was confirmed and documented. (#396) The proton was moved onto a *trivalent junction* W_{ABC}, where we: built a valid-manifold junction that conserves color charge and respects S_3; fixed a relaxation stall; showed that a van Raamsdonk metric from *pairwise* mutual information cannot select the singlet; realized the photon as a null worldline under a Lorentzian relax; and — the headline result — proved that the singlet is reachable by the input→result *transport* (it lies in the image of that linear map), even though the actual *merge with a natural input does not reach it* ($\sim 74\%$). The ceiling is a suboptimal input on gauge-asymmetric windows, not a geometry limit; the one remaining task is to produce the singlet from a *principled* input, via a symmetric (gauge-clean) window placement.

1 The goal and the method

A proton is a color-singlet bound state of three quarks. In this subsystem matter is not imposed: color states are pinned on the boundary ∂W of a Regge cobordism W, the interior edge lengths relax to a stationary point of the dual (Lorentzian) Regge action, and the emergent result is *read off* the relaxed geometry. Color lives in the homology of a holed register surface: three holes carry the three colors, the S_3 permuting them is the (Weyl) gauge group, and a configuration is color-neutral iff its oriented charge $\Sigma(\psi) = \sum_i \psi_i$ vanishes. The proton target is the $\mathbb{Z}_3$ definite-triality singlet $[1, \omega, \omega^2]$, $\omega = e^{2\pi i/3}$.

2 #382 — the bipartite obstruction

The theorem. The $SU(3)$ color singlet is built from the invariant tensors $\delta^i{}_j$ (a meson, $\mathbf{3} \otimes \bar{\mathbf{3}} \supset \mathbf{1}$) and ε_{ijk} (a baryon, three quarks). Crucially $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ contains *no* singlet: the two-quark Hilbert space has no colorless state. The singlet first appears in $\mathbf{3}^{\otimes 3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$. Hence a baryon singlet is irreducibly three-body, and no single bipartite fusion of two color triplets can produce it. The qudit form is sharp — baryon arity $= d$ for $SU(d)$, so a *qubit* baryon *is* bipartite (ε_{ij}, the spin singlet) while a *qutrit* color baryon is not.

What we did. We refactored the register `MergeCobordism` onto the *exact* period read-out (`residualForPeriods/cyclePeriods`, pinned inputs only, emergent result), removing the soft edge-loop encoding so the register cannot silently fall back to it. The composed bipartite *sequence* ($A, B \to AB$, then $AB, C \to ABC$) stays color-charged at every step ($|\Sigma| \approx 0.67$), exactly reproducing the obstruction. The argument is written up in `proton_bipartite_obstruction.pdf`. The conclusion — the proton needs a *tripartite* junction — motivated #396.

3 #396 — the trivalent junction W_{ABC}

The junction is one connected base surface (a 2-frequency geodesic icosahedron, S^2, 42 vertices) minus 12 vertex-disjoint hole triangles grouped into **four windows of three** — A, B, C (inputs) and R (the emergent result) — extruded $\times I$. It is a valid manifold ($b_1 = 11$, `dualComplexValid`, no weld), the implementation being `TripartiteRegisterTopology`.

3.1 Distinct holes, not stacked layers

A single *shared* register averages its inputs (one period asked to equal three targets $\to$ least-squares mean $\to$ the symmetric, non-singlet part); both the $b_1 = 2$ linear staircase and the $b_1 = 3$ four-holed sphere stack the inputs on one shared cycle and fail the same way. The fix is *distinct holes* — independent 1-cycles — where the global Stokes relation makes confinement a *conservation law at the junction*,

$$\sum_{\text{all holes}} (\text{induced period}) = 0 \implies \Sigma_R = -\Sigma_{\text{inputs}}.$$

Measured: the three neutral pairs carry *exactly* ($r_U \sim 10^{-27}$); the emergent result is neutral for neutral inputs and carries the net charge for colored inputs; the construction is S_3-respecting.

3.2 The relaxation fix

The junction relaxation initially stalled at iteration 0. Because the inputs are carried exactly on independent cycles, $r_{\text{state}} \sim 10^{-27}$ on any metric, so its analytic gradient is numerical noise; folded through the ill-conditioned action Gauss–Newton Hessian it exploded the step ($\|\delta\| \sim 10^{13}$). The fix gates the state-residual gradient — folding it in only while r_{state} exceeds its convergence floor — and guards the line search against degenerate-step exceptions. The junction now relaxes ($0 \to 40^+$ iterations, $\|\nabla S\|^2 : 315 \to 14$); the bipartite/operator paths ($r_{\text{state}} \gg \varepsilon$) are unchanged.

3.3 Van Raamsdonk from entanglement — a sharp negative result

"Geometry from entanglement" suggests seeding the metric $\ell^2 = (-\log(I/I_{\max}))^2$ from the singlet's mutual information. It cannot select the singlet: for *both* the antisymmetric ε state and the $\mathbb{Z}_3$ clock state, every two-party reduced density matrix has spectrum $\{0, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$, so $S(\rho_X) = I(X{:}Y) = \log 3$ — the *pairwise* mutual information is maximal and *uniform.* The singlet's content is a three-party correlation, invisible to any pairwise marginal (the same irreducible-three-body fact as the ε obstruction), and the VR length is real, so it cannot carry a phase. Empirically the VR metric does not move the result (0.737 vs. the 0.744 jitter baseline).

3.4 Lorentzian geometry and the photon

The ω-phases of $[1, \omega, \omega^2]$ are complex; a real metric gives real harmonics that cannot represent them. Making the forward-time worldline edges timelike ($\ell^2 < 0$) takes the dual Regge action complex ($\operatorname{Im} S \neq 0$), so its harmonics can. The same lever realizes the photon: relaxing the Lorentzian junction drives a worldline edge through *null* ($\ell^2 \to 0$, lightlike) — the predicted photon, an edge going null during the interaction.

3.5 The headline: the ceiling is the input, not the geometry

Across jitter, VR and Lorentzian metrics the singlet-overlap held at ~ 0.74, suggesting a structural ceiling. It is not. The carried-input read is *linear* in the inputs, so the input→result transport is a matrix $M \in \mathbb{C}^{3\times 9}$. We find $\operatorname{rank} M = 3$ and $[1, \omega, \omega^2] \in \operatorname{im} M$:

$$\max_{\text{neutral inputs } t} \frac{|\langle Mt,\ [1,\omega,\omega^2]\rangle|}{\|Mt\|\,\|[1,\omega,\omega^2]\|} = 1.0000.$$

The exact singlet is *reachable by the transport* — $\exists$ a neutral input whose merge produces it — so the 74% was a suboptimal *input* encoding, not a limit of the junction.

Transport-reachable vs. merge-produced (the honest distinction). "Reachable" here means the singlet is in the *image* of M: some input maps to it. It does *not* mean the merge *produced* the singlet — run with a *natural* (physical, symmetric) quark input the merge still gives ~ 0.74. The only input that yields the singlet is $M^{-1}[1, \omega, \omega^2]$, which is *geometry-specific* (reverse-engineered from the relaxed metric), not an independently motivated state. So the result is: the junction *can* carry the proton (necessary), but we have not yet *produced* it from a principled input through the merge (sufficient). Closing that gap is exactly the symmetric-window task below — with gauge-clean windows the natural input *is* the matched input, and the merge produces the singlet.

3.6 The gauge structure is geometric

Is the singlet-producing input simply the gauge-rotated symmetric singlet? No. The per-window transport blocks satisfy that $M_A^{-1} M_B$ and $M_A^{-1} M_C$ are *general* matrices, not signed permutations — so the windows are geometrically inequivalent (greedy placement), not related by an S_3 color relabeling. Natural *symmetric* inputs therefore cap at 0.70–0.81; only a geometry-specific (reverse-engineered) input reaches 1.0. A *principled* natural-input→singlet requires symmetric (tetrahedral) window placement.

4 Status and next step

Established	**State**
Bipartite cannot make the singlet (#382)	proved + documented; PR merged
Trivalent junction (valid manifold, charge-conserving, S_3)	built (`TripartiteRegisterTopology`)
Relaxation fix	landed; junction relaxes
VR-from-pairwise-MI is uniform	negative result, documented
Photon (null worldline)	realized under Lorentzian relax
Singlet reachable by the *transport*	proved ($[1, \omega, \omega^2] \in \operatorname{im} M$, max overlap 1.0)
. . . but *not* produced by a natural-input merge	merge gives $\sim$ 0.74; the matched input is geometry-specific
Window asymmetry is geometric, not gauge	proved ⇒ symmetric windows needed

The proton singlet is *achievable* on the junction. The single remaining piece is symmetric window placement: four windows at tetrahedrally-symmetric positions, each a C_3-symmetric hole triple with a consistent induced-sign pattern, so a natural symmetric quark input transports to the exact singlet with manifest S_3 symmetry, and the photon emerges from the Lorentzian worldlines.

Artifacts. `TripartiteRegisterTopology` (build, exact read-out, `set_entangled_metric`, `set_lorentzian_wor` the `MergeCobordism` relaxation fix and emergent-read getters; and three design notes — `proton_bipartite_obstru` (why bipartite fails), `proton_tripartite_junction_choices.pdf` (the four design choices), and this review.